# A120596 concise calculus certificate

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## Definition and first terms

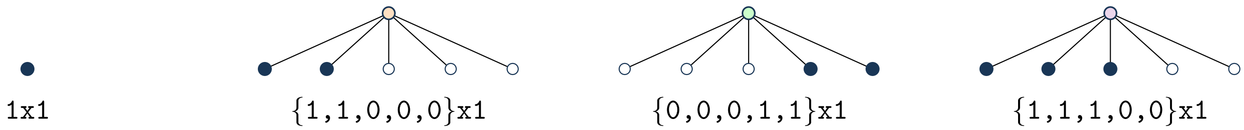
$6\,A(x) = 5 + 1\,x + A(x)^5$, with the origin-normalized solution.

Normalization/grammar: `A(x)=1+(1)*T(x); T=x+10*T^2+10*T^3+5*T^4+1*T^5.`

$a(0), \ldots, a(7) = 1, 1, 10, 210, 5505, 161601, 5082420, 167451780.$

Kernel used throughout: $\rho(u) = u\,(1 - 10\,u^1 - 10\,u^2 - 5\,u^3 - 1\,u^4)$, so $x = \rho(T(x))$.

## Typogeometry




literal 5-slot geometry; empty positions are shown. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: $\Delta_2\colon\ 10,\ \Delta_3\colon\ 10,\ \Delta_4\colon\ 5,\ \Delta_5\colon\ 1.$

## Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \ldots, k_4) \in \mathbb{Z}^4_{\geq 0} : \sum_{j=1}^{4} j k_j = n - 1\}, \quad K = \sum_{j=1}^{4} k_j,$$

$$a(n) = \frac{1}{n} \sum_{\boldsymbol{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \ldots, k_4} 10^{k_1} 10^{k_2} 5^{k_3} 1^{k_4}.$$

$$a(n) = \frac{1}{2\pi i n} \oint_\gamma \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

## Recurrence

$\sum_{r=0}^{4} P_r(n) a(n+r) = 0$, $n \geq 1$, with

$$P_0(n) = -3125\,n^4 - 12500\,n^3 - 13750\,n^2 - 2500\,n + 1155$$
$$P_1(n) = -62500\,n^4 - 343750\,n^3 - 668750\,n^2 - 546875\,n - 159375$$
$$P_2(n) = -468750\,n^4 - 3281250\,n^3 - 8390625\,n^2 - 9234375\,n - 3656250$$
$$P_3(n) = -1562500\,n^4 - 13281250\,n^3 - 40625000\,n^2 - 52343750\,n - 23437500$$
$$P_4(n) = 37531\,n^4 + 375310\,n^3 + 1313585\,n^2 + 1876550\,n + 900744$$

## Differential equation

The scalar linear ODE has order 4. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `case.json#/objects/ode_from_recurrence`.

## Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

### Certificate.

$R(n,u) = N(n,u)/\rho(u)^3,\ \deg_u N = 16.$

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n,u) H_n(u))\,.$

Matrix dimensions: $10 \times 10$; remainder $4 \times 5$, rank 4, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Canonical records: `data/tree_model.json`, `data/contour.json`, `data/matrices.json`, `data/recurrence.json`, `data/ode.json`. Embedded payload contains exact sources and SHA-256 matrix identifiers. Compare with the verbose certificate for A120590.